

Magneto-Optical Induction Apparatus (MOIA)

Complete Consolidated Research Package — Evidence Audit, Physics Assessment, Validation Program, Applications, Publication and IP Strategy

Vanthrix Research | 16 August 2026

Status: consolidated scientific research record. Established principles, observations, hypotheses, and unsupported claims are explicitly separated.

Executive Summary

MOIA is best treated as a multiphysics experimental platform integrating electromagnetic excitation, optical illumination and polarization, photothermal processes, mechanical elements, and camera/display imaging. The available project record supports several strong, conventional-physics components and one important unresolved coupling.

The strongest documented research direction is the camera/display imaging chain. A recorded signal is displayed and then re-sampled by a second camera. Changes in geometry, motion, display conditions, and optical elements were reported to produce changes in color, brightness, geometry, apparent flattening, inversion, banding, and other image structures. These observations are strongly compatible with established effects including spatial and temporal aliasing, Moiré, rolling-shutter distortion, polarization, birefringence, and camera/display processing.

The most important unresolved phenomenon is the reported correlation between blue illumination, a dark object, and nearby electrical/coil geometry with apparent mechanical or state changes. The current archive does not contain enough quantitative electrical, magnetic, thermal, mechanical, and optical measurements to identify the mechanism. This must remain an open hypothesis rather than a claim of a new force.

The project should therefore be presented in three tracks: Track A, computational imaging and camera/display physics; Track B, electromagnetic/photothermal/mechanical coupling; and Track C, physical-layer optical encoding. The existing evidence is sufficient to produce a serious research record and replication framework without asserting new physics.

1. Research Identity and Scope

Working technical identity: a multiphysics experimental platform for investigating physical optical-state modulation and camera–display signal transformations through coupled electromagnetic, optical, thermal, mechanical, and digital-imaging processes.

Recommended public description:

“MOIA is an experimental platform for studying coupled electromagnetic, optical, thermal, mechanical, and camera–display phenomena, with emphasis on polarization states, optical modulation, and physical imaging transformations.”

Do not define MOIA primarily as a black-hole analogue, new-force apparatus, or encryption device. Those concepts may remain as conceptual hypotheses or application directions, but they are not presently established conclusions.

2. Project Provenance and Evidence Boundary

The consolidated archive includes the Multiphysics Optical Encoding Platform report, optical/electromagnetic mechanism reports, executive summaries, experiment summaries, Zenodo preparation material, and historical research notes. Some original videos and photographs referenced by the notes are not presently available for direct frame-by-frame or metadata analysis.

Accordingly, this package uses five evidence labels:

ESTABLISHED — follows from established physics and is applicable to the relevant apparatus.

OBSERVED/REPORTED — described or documented by the experimenter, but not currently independently reprocessed from complete raw data.

PLAUSIBLE — a known physical mechanism that can reasonably explain an observation but has not been quantitatively isolated in the apparatus.

UNRESOLVED/HYPOTHESIS — competing mechanisms remain possible and measurements are required.

UNSUPPORTED — the available evidence does not justify the claim.

This distinction is mandatory for all future public material.

3. Apparatus Architecture

The reconstructed apparatus contains combinations of:

- electrical conductors/coils and AC excitation;
- dark/black conductive or magnetic/mechanical objects;
- blue LED illumination;
- water/glass optical elements;
- prescription lenses and polarization interactions;
- phone cameras and displays;
- recorded-video playback and second-stage imaging.

Conceptual architecture:

EM subsystem → field/mechanical environment

Optical subsystem → illumination/polarization/transmission

Thermal subsystem → absorption/heating/environmental gradients

Mechanical subsystem → rotation/displacement/vibration

Imaging subsystem → sampling, display feedback, aliasing and digital processing

The central engineering idea is integration, not the assumption that every subsystem necessarily drives every other subsystem.

4. Master Observation Record

A. Camera/display experiment.

A previously recorded video was displayed and then recorded again. Changes in camera orientation, display geometry, motion, and optical conditions were reported to produce large changes in apparent color, brightness, geometry, motion, and dimensional appearance. Apparent flattening and inversion were also reported.

B. Polarization and lens experiment.

Crossed-polarizer states and prescription lenses produced strong visual intensity/color structures. Stress birefringence is a well-established candidate mechanism for structured changes when stressed transparent lenses are placed between polarization-selective elements.

C. Blue-light/electrical-geometry observation.

Blue illumination was associated with a dark object in a setup containing bent/coiled electrical geometry. Apparent motion or activation was reported. The current record lacks sufficient simultaneous measurements to determine whether the effect was magnetic, thermal, optical, mechanical, or a coupled combination.

D. Water/glass optical setup.

A blue LED was placed beneath a water-filled wine glass, with a paper tube used to shape the beam and a small circular object/keychain in the optical environment. The defensible interpretation is an optical/refraction/illumination experiment unless future measurements establish another mechanism.

5. Physics Foundations and Equations

5.1 Polarization.

For ideal linear polarizers:

$$I = I_0 \cos^2(\theta).$$

This gives the basis for optical extinction and modulation. Real systems have finite extinction ratios, scattering, leakage, and wavelength dependence.

5.2 Electromagnetic induction.

Faraday's law:

$$\text{EMF} = -d(\Phi_B)/dt.$$

AC current creates a time-varying magnetic field. Mechanical torque depends on field geometry and the magnetic/electrical properties of the target.

5.3 Magnetic torque.

For a magnetic dipole approximation:

$$\tau = \mathbf{m} \times \mathbf{B}.$$

This approximation is only valid when the target can reasonably be modeled by an effective magnetic moment. The available archive does not provide the measured parameters needed to calculate the actual apparatus torque.

5.4 Photothermal response.

A first-order temperature estimate is:

$$\Delta T \approx P_{\text{abs}} t / (m c),$$

before accounting for conduction, convection, radiation, geometry, and spatially localized heating.

5.5 Faraday rotation.

In a magneto-optic medium, the polarization rotation may be represented approximately as:

$$\theta_F = V B L,$$

where V is the Verdet constant, B is magnetic field, and L is optical path length. A normal copper conductor/coil is not by itself a strong Faraday rotator; a suitable magneto-optic medium is generally required.

5.6 Imaging and sampling.

A camera recording a display samples an already sampled spatial/temporal signal. The display contributes pixel/subpixel structure and refresh timing; the camera contributes pixel sampling, exposure, frame rate, rolling shutter, optical filtering, and processing. Their interaction can create:

- spatial aliasing;
- temporal aliasing;
- Moiré;
- rolling-shutter skew/banding;
- color-channel artifacts;
- apparent geometric transformations.

These mechanisms provide the strongest conventional-physics explanation for the camera/display portion of the project.

6. Claim-by-Claim Evidence Matrix

1. Crossed-polarizer optical extinction — ESTABLISHED.
2. Malus-law intensity dependence — ESTABLISHED.
3. Stress birefringence in transparent lenses — ESTABLISHED mechanism.
4. Display/camera Moiré — ESTABLISHED mechanism.
5. Rolling-shutter distortion — ESTABLISHED mechanism.
6. Blue-light absorption by dark surfaces — ESTABLISHED.
7. Coil-induced magnetic field — ESTABLISHED in principle; apparatus field not measured in the available record.
8. Blue-light/dark-object/electrical-geometry correlation — OBSERVED/REPORTED.
9. Mechanism of that correlation — UNRESOLVED.
10. Blue light acting as a magnetic switch — UNSUPPORTED.
11. New force — UNSUPPORTED.
12. Literal black-hole analogue — UNSUPPORTED physically; acceptable only as conceptual analogy.
13. Secure encryption — UNSUPPORTED.
14. Physical-layer optical encoding concept — PLAUSIBLE APPLICATION.
15. Multiphysics experimental platform — SUPPORTED engineering description.

7. Central Unresolved Research Question

The central unresolved question should be stated as:

“Does the specific blue-light/electromagnetic configuration produce a repeatable mechanical, optical, electrical, or thermal response that cannot be explained by independently measured conventional mechanisms?”

This formulation is intentionally conservative. It does not assume a new force. It creates a falsifiable program.

Required simultaneous measurements:

- LED optical power at the target;
- wavelength and illumination geometry;
- object temperature versus time;
- object displacement/velocity versus time;

- coil voltage/current;
- AC frequency and waveform;
- magnetic field at the object;
- object mass, material and dimensions;
- object-to-coil distance and orientation;
- synchronized video;
- ambient temperature and airflow where relevant.

8. Validation Program

Experiment A — Polarization calibration.

Measure transmitted intensity versus polarizer angle. Fit the data to Malus's law plus a finite background/extinction term. Repeat with the prescription lens inserted and rotated.

Experiment B — Camera/display characterization.

Display standardized test patterns. Record at controlled shutter speeds, frame rates, distances, angles, focus settings and display refresh conditions. Extract line profiles and FFT spectra to quantify Moiré and banding.

Experiment C — Blue-light thermal control.

Measure target temperature with blue illumination on/off and with an equal-power control source if possible. Record displacement simultaneously. Compare observed motion with thermal predictions.

Experiment D — Magnetic-field mapping.

Measure the field at the target location with the coil energized/de-energized. Record current and frequency. Map B versus position.

Experiment E — Mechanical artifact controls.

Repeat with light blocked, coil unpowered, target replaced, airflow minimized, and mechanical supports changed. The goal is to isolate vibration, convection, heating, magnetic interaction, and optical effects.

Experiment F — Full multiphysics run.

After subsystem calibration, activate the system together while synchronizing electrical, optical, thermal, mechanical, and camera measurements.

9. Quantitative Analysis Pipeline

For every experiment, preserve raw data before processing.

Recommended pipeline:

1. Raw acquisition.
2. Metadata capture.
3. Dark/zero calibration.
4. Baseline/control subtraction where justified.
5. Time synchronization.
6. Region-of-interest extraction.
7. Spatial/temporal filtering only after raw data are preserved.
8. FFT/spectral analysis for imaging patterns.
9. Regression against controlled independent variables.
10. Uncertainty propagation.
11. Repeatability analysis.
12. Comparison against conventional physical models.
13. Residual analysis.
14. Independent replication.

For the camera/display track, quantify pattern frequency, luminance, RGB response, geometric displacement, and frame-to-frame motion.

For the EM/thermal track, quantify field, current, temperature, force/displacement, and time delay.

10. Pass/Fail Criteria

Camera/display hypothesis:

PASS if measured visual structures change predictably with controlled display/camera sampling variables and can be reproduced.
FAIL if the claimed transformation persists after the relevant sampling mechanisms are suppressed.

Thermal/mechanical hypothesis:

PASS for an anomalous coupling only if the measured mechanical response exceeds combined measurement uncertainty, is repeatable, survives controls, and cannot be accounted for by independently bounded magnetic, thermal, airflow, vibration, or optical effects.

Otherwise classify it as conventional, unresolved, or artifact depending on the evidence.

Polarization hypothesis:

PASS if measured transmission follows the expected angular dependence and the lens introduces a reproducible birefringent modulation.

11. Applications — Full Field Map

1. Computational imaging and camera/display characterization — strongest near-term scientific application.
2. Optical engineering — polarization control, birefringence, extinction and transmission.
3. Physical-layer optical encoding — represent information through controlled physical optical states.
4. Sensor/display diagnostics — use standardized patterns to characterize sampling interactions.
5. Optical communications research — investigate modulation and receiver behavior; not yet a secure communication system.
6. Interactive media and kinetic art — real-time physical optical transformations.
7. STEM education — integrated demonstrations of electromagnetism, optics, thermal physics and imaging.
8. Experimental photonics — test controlled coupling between optical and electromagnetic subsystems.
9. Computational photography — study how re-imaging a displayed signal transforms information.
10. Optical authentication/anti-counterfeit research — potential use of polarization-dependent physical states, subject to security testing.

12. Application Ranking by Evidence

Tier 1 — immediately defensible:

- computational imaging;
- camera/display characterization;
- optical engineering demonstration;
- STEM education;
- interactive optical media.

Tier 2 — technically plausible but requires additional engineering:

- physical-layer optical encoding;
- optical authentication;
- optical communications experiments;
- sensor calibration/diagnostics.

Tier 3 — speculative until validated:

- anomalous EM/photonic coupling;
- new physical interaction;
- black-hole physical analogue;
- cryptographic security claims.

13. Publication Package

Recommended public package:

File 1 — Master technical report.
This document.

File 2 — Camera/display replication protocol.

A concise experimental procedure with diagrams, test-pattern requirements, camera settings, and analysis methodology.

File 3 — Experimental observation archive.

Available original photographs, videos, screenshots, apparatus photographs, and annotated figures.

File 4 — Analysis notebook.

Python/MATLAB scripts for FFT analysis, image registration, line profiles, color-channel analysis, and motion extraction.

File 5 — Evidence matrix.

A one-page claim/status table.

File 6 — Research README.

Explains what is included, what is missing, and how to reproduce the strongest experiments.

The existing Zenodo record should remain the foundational public record. The next release should be narrower and more reproducible, emphasizing the camera/display work and clearly labeling the unresolved EM/photothermal coupling.

14. IP and Disclosure Strategy

Scientific publication and invention disclosure should be separated.

Public record:

- observations;
- established physics;
- apparatus overview;
- replication methodology;
- data that can safely be released;
- limitations.

Private invention disclosure:

- potentially novel geometry;
- specific control sequence;
- implementation details;
- proposed commercial embodiment;
- alternative embodiments;
- dates and provenance;
- drawings and claim concepts.

Do not assume that a public research upload proves patentability. Patentability is a legal determination. Any potentially valuable implementation should be reviewed by qualified patent counsel before additional enabling disclosure.

15. Black-Hole / Cosmic Analogy — Proper Scientific Treatment

The black-hole discussion can remain in the project as a conceptual analogy involving rotation, magnetic fields, plasma, polarization, and energy flow. It should not be used as evidence that MOIA reproduces black-hole physics.

Real rotating black-hole systems involve general relativity, relativistic plasma, strong gravity, and astrophysical magnetic fields. Laboratory optical and electromagnetic systems can share mathematical or conceptual motifs without being physically equivalent.

The appropriate wording is:

“MOIA provides a laboratory-scale conceptual analogy for coupled rotation, electromagnetic fields, and optical information flow; it is not a physical simulation of a black hole.”

16. Historical Matrix

1845 — Faraday effect: magnetic-field-dependent optical polarization rotation.

1887 — Tesla's Egg of Columbus: rotating magnetic fields and induced mechanical rotation.

Late 19th/early 20th century — photophoretic and radiation-pressure research.

20th century — polarization optics and optical isolation.

Modern imaging era — CMOS sampling, rolling shutter, digital color processing, and display/camera aliasing.

MOIA does not claim to rediscover these phenomena. Its research value is the deliberate experimental integration and the resulting cross-domain measurement questions.

17. Limitations

- Original raw media are incomplete in the current accessible archive.
- Exact electrical parameters are not consistently available.
- Magnetic-field measurements are absent from the existing record.
- Optical power and thermal measurements are absent.
- Object material and mechanical properties are incompletely characterized.
- Some historical descriptions use qualitative language rather than instrument measurements.
- Some earlier calculations used illustrative assumptions; these must not be represented as measured values.
- Camera automatic exposure, white balance, sharpening, HDR, stabilization, and other processing may influence recorded results.
- No cryptographic security analysis has been performed.
- No independent laboratory replication has yet been established.

18. What Counts as Success

The project does not need to prove new physics to be successful.

Success at the first level is a reproducible demonstration of camera/display transformations under controlled conditions.

Success at the second level is a quantitative multiphysics platform that predicts and controls optical output from known physical inputs.

Success at the third level would be a repeatable residual effect that survives controls and cannot be explained by the established mechanisms tested.

Only at that third level would extraordinary physical claims become appropriate for serious consideration.

19. Final Scientific Position

MOIA should now be presented as a multiphysics experimental research platform.

The strongest part is the optical/camera/display system, where conventional imaging physics provides a rigorous framework and a reproducible path.

The most interesting unresolved part is the blue-light/electrical-geometry/mechanical correlation. It deserves controlled measurement, not premature interpretation.

The practical applications should begin with computational imaging, optical engineering, education, sensor/display characterization, and interactive optical media. Physical-layer encoding and optical authentication are plausible engineering directions that require additional validation. Cryptography, new forces, and literal black-hole physics remain outside the demonstrated evidence.

The project is therefore ready for professional consolidation, controlled replication, scientific outreach, and separate IP review. The correct next stage is no longer broad speculation; it is measurement, reproducibility, and disciplined dissemination.

Appendix A — Rapid Evidence Matrix

- E1 — Polarization extinction: ESTABLISHED.
- E2 — Malus-law dependence: ESTABLISHED.
- E3 — Lens stress birefringence: ESTABLISHED mechanism; apparatus-specific contribution pending.
- E4 — Display/camera Moiré: ESTABLISHED mechanism.
- E5 — Rolling-shutter distortion: ESTABLISHED mechanism.
- E6 — Blue-light absorption/heating: ESTABLISHED.
- E7 — Coil magnetic field: ESTABLISHED in principle; apparatus field unmeasured.
- E8 — Blue-light/object/electrical-geometry correlation: OBSERVED/REPORTED.
- E9 — Mechanism of correlation: UNRESOLVED.
- E10 — New force: UNSUPPORTED.
- E11 — Black-hole physical equivalence: UNSUPPORTED.
- E12 — Secure encryption: UNSUPPORTED.
- E13 — Physical optical encoding concept: PLAUSIBLE.
- E14 — Multiphysics platform architecture: SUPPORTED.

Appendix B — Replication Data Sheet

Record before each run:
Date/time; operator; apparatus configuration; object material/mass/dimensions; coil geometry; current; voltage; frequency; magnetic field; LED wavelength; optical power; beam diameter; target distance; ambient temperature; target temperature; camera model; lens/focus; shutter; frame rate; ISO/exposure; display model; display refresh rate; display brightness; camera-display distance; camera angle; polarization angle; notes.

Every run should have a unique identifier. Preserve the raw file before any editing.

Appendix C — Public Abstract

MOIA is a multiphysics experimental platform developed to investigate interactions among electromagnetic excitation, optical illumination and polarization, thermal processes, mechanical elements, and digital imaging. The platform includes optical-state modulation and camera–display re-imaging experiments in which a displayed signal is sampled by a second camera. The resulting transformations include changes in intensity, color, geometric appearance, Moiré, banding, and apparent flattening or inversion. These effects are compatible with established polarization, birefringence, spatial/temporal sampling, rolling-shutter, and display/camera interaction mechanisms. A separate reported correlation between blue illumination, a dark object, and nearby electrical geometry remains unresolved because the available record lacks synchronized electrical, magnetic, thermal, and mechanical measurements. The work is therefore presented as an experimental platform and research program rather than as evidence of new physics or a validated cryptographic system. The principal contribution is the integration of multiple physical domains into a testable architecture and the identification of controlled experiments capable of separating conventional mechanisms from any residual anomalous response.

Appendix D — Suggested Short Title and Keywords

Short title: MOIA — Multiphysics Optical Imaging and Induction Platform

Keywords:
magneto-optics; polarization optics; optical encoding; computational imaging; camera-display interference; Moiré; rolling shutter; photothermal effects; electromagnetic induction; optical modulation; physical-layer signaling; sensor characterization; experimental photonics; multiphysics systems.

Appendix E — Claim Status Table

Claim	Status	Treatment
Polarization extinction	Established	Use as physics basis.
Camera/display Moiré & rolling shutter	Established mechanism	Primary reproducible track.
Lens birefringence	Established mechanism	Verify quantitatively.
Blue-light/electrical/mechanical coupling	Unresolved	Open hypothesis; measure.
New force	Unsupported	Do not claim.
Black-hole physical analogue	Unsupported physically	Conceptual analogy only.
Secure encryption	Unvalidated	Do not call cryptographic security.
Multiphysics platform	Supported	Central engineering description.